

Physics Simulation and STRidge Identification Report

Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.

1. Simulation Parameters

Quantity	Symbol	Value	Unit
Fluid density	ρ	1.00000000e+03	kg m ⁻³
Dynamic viscosity	ν	1.00000000e+01	Pa s
Kinematic viscosity	ν	1.00000000e-02	m ² s ⁻¹
Characteristic velocity	$u_{\max}$	1.00000000e+00	m s ⁻¹
Characteristic length	D	1.00000000e+00	m
Reynolds number	Re	1.00000000e+02	dimensionless
Body force	G	9.81000000e+00	m s ⁻²
Grid points in x	N_x	3.10000000e+01	cells
Grid points in y	N_y	3.10000000e+01	cells
Grid spacing in x	Δx	3.22580645e-02	m
Grid spacing in y	Δy	3.22580645e-02	m
Time step	Δt	1.00000000e-04	s
Numerical threshold	ϵ	7.00000000e-11	-

2. Numerical Discretisation

Quantity	Symbol	Value	Unit
Parameters listed above are the values used by the selected solver.			

3. Data and Pre-processing

Quantity	Value
Loaded field shape (t, x, y)	(6613, 80, 80)
Regression field shape (t, x, y)	(4613, 76, 76)
Regression samples	26,644,688
Dictionary terms	5
First saved time index	150

Quantity	Value
Temporal crop at each boundary	1000 layers
x-boundary crop	2 points
y-boundary crop	2 points

4. Governing Equation Verification

Reference equation

$$u_t = -1.00000000e+00*uu_x - 1.00000000e+00*vu_y + 1.00000000e-02*u_{xx} + 1.00000000e-02*u_{yy} - 1.00000000e-03*p_x$$

Learned equation

$$u_t = -1.00147275e+00*uu_x - 1.02149164e+00*vu_y + 7.95481353e-03*u_{xx} + 9.87097214e-03*u_{yy} - 1.01326174e-03*p_x$$

Term	Reference coefficient	Learned coefficient	Percentage error
uu_x	-1.00000000e+00	-1.00147275e+00	-1.47275300e-01%
vu_y	-1.00000000e+00	-1.02149164e+00	-2.14916421e+00%
u_xx	+1.00000000e-02	+7.95481353e-03	2.04518647e+01%
u_yy	+1.00000000e-02	+9.87097214e-03	1.29027861e+00%
p_x	-1.00000000e-03	-1.01326174e-03	-1.32617389e+00%

5. STRidge Selection and Error Metrics

Metric	Value
Regularisation parameter, λ	1.00000000e-08
Selected tolerance	9.42668455e-04
L0 penalty	1.00000000e-06
Training RMSE	1.82186069e-02
Validation RMSE	2.21005709e-02

6. Reproducibility Notes

- Input directory:
/Users/minnie/Desktop/PhysicsFYP/ns_solver/stridge_model/output/lidcavity_81
- Fields were loaded starting at saved time index 150.
- Derivatives were calculated using second-order edge-aware finite differences via `numpy.gradient`.
- The learned equation is compared term-by-term with the supplied reference coefficients.

7. Transverse Momentum Equation

Reference equation

$$v_t = -1.00000000e+00*uv_x - 1.00000000e+00*vv_y + 1.00000000e-02*v_{xx} + 1.00000000e-02*v_{yy} - 1.00000000e-03*p_y$$

Learned equation

$$v_t = -1.25826371e+00*uv_x - 9.77523408e-01*vv_y + 8.91606257e-03*v_{xx} + 1.24131022e-02*v_{yy} - 9.99830985e-04*p_y$$

Term	Reference coefficient	Learned coefficient	Percentage error
uv_x	-1.00000000e+00	-1.25826371e+00	-2.58263715e+01%
vv_y	-1.00000000e+00	-9.77523408e-01	-2.24765922e+00%
v_xx	+1.00000000e-02	+8.91606257e-03	1.08393743e+01%
v_yy	+1.00000000e-02	+1.24131022e-02	2.41310223e+01%
p_y	-1.00000000e-03	-9.99830985e-04	-1.69015328e-02%

V STRidge Selection and Error Metrics

Metric	Value
Selected tolerance	9.42668455e-04
Training RMSE	9.25348313e-03
Validation RMSE	1.21935714e-02